Greedy Algorithm Visualizations

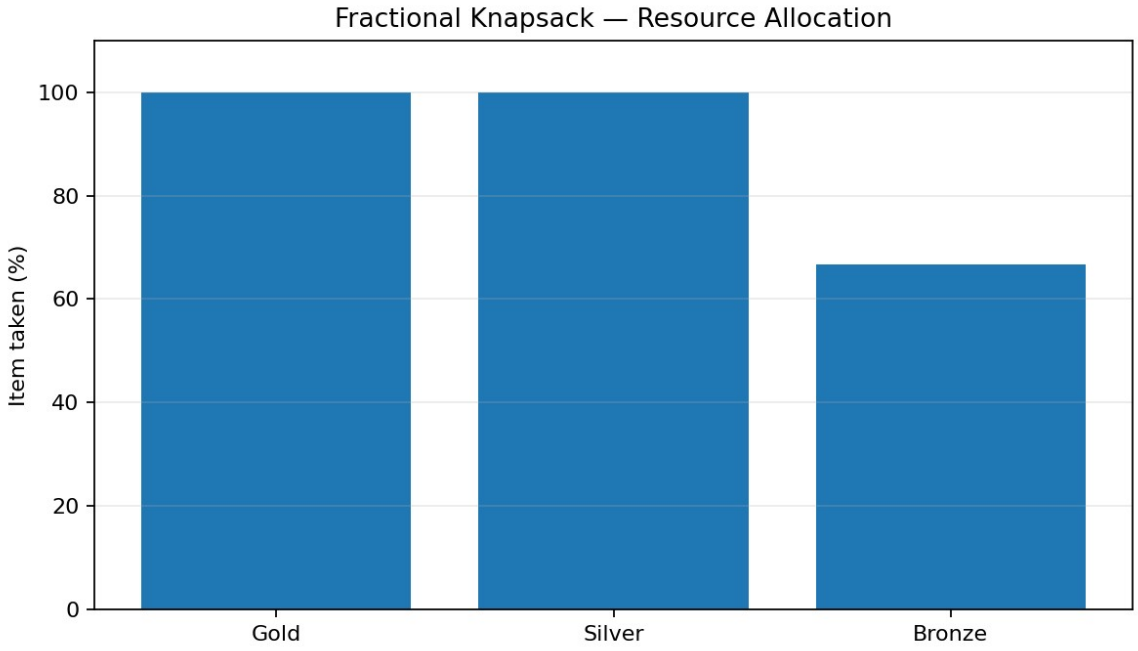
Lab 3 — Resource Allocation, Scheduling, Huffman Tree and Performance

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Lab Name: Lab 3 — Smart Resource Allocation and Scheduling System using Greedy Algorithms

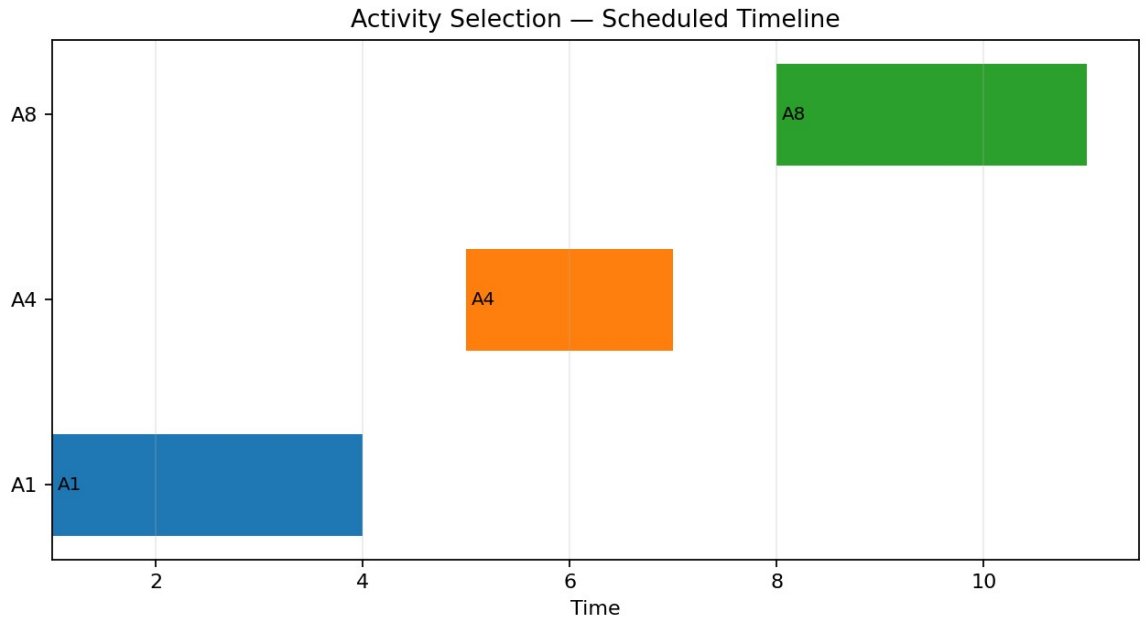
1. Fractional Knapsack — Resource Allocation

The chart shows the fraction of each selected item placed into a capacity-50 knapsack. Gold and Silver are fully selected; two-thirds of Bronze is selected. Total value = 240.



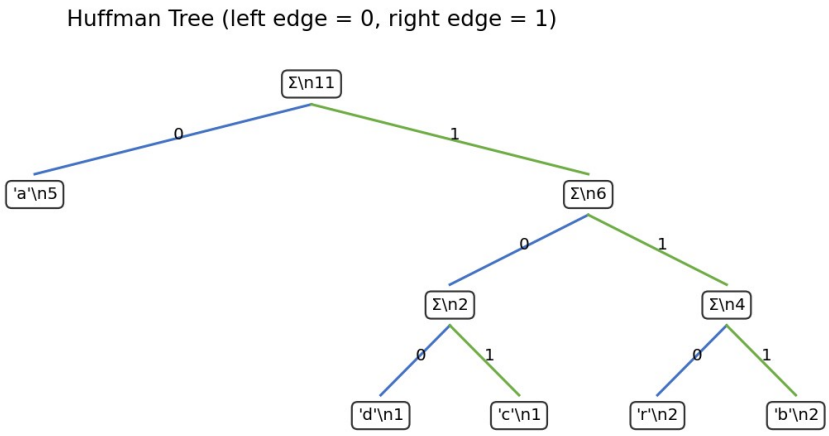
2. Activity Selection — Timeline

The earliest-finish-time greedy strategy selects A1 [1,4], A4 [5,7], and A8 [8,11], producing three non-overlapping activities.

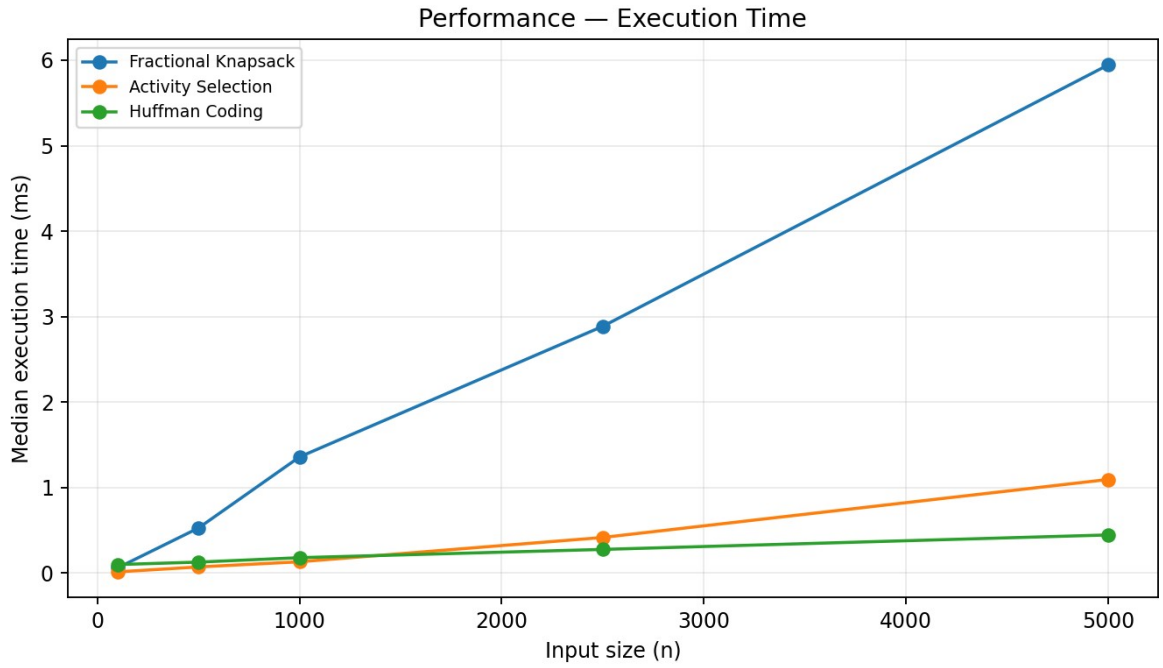


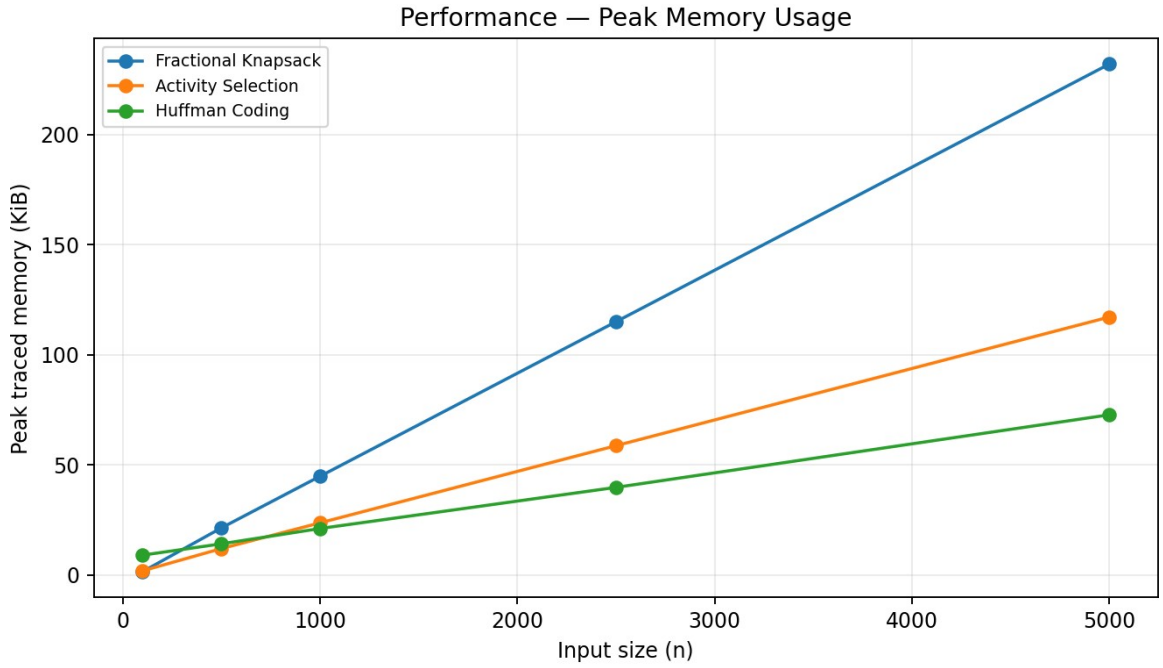
3. Huffman Tree

Huffman Coding repeatedly merges the two least-frequent nodes. Left edges represent 0 and right edges represent 1. The displayed tree is generated from the sample string 'abracadabra'.



4. Performance Graphs





5. Program Output Screenshot

Generated from the actual execution of greedy_lab3.py.

```
Windows Terminal  |  Python  |  Lab 3

> python greedy_lab3.py


LAB 3: SMART RESOURCE ALLOCATION AND SCHEDULING
==================================================
Knapsack maximum value: 240.00
Allocation: Gold=100.0%, Silver=100.0%, Bronze=66.7%
Selected activities: A1[1,4], A4[5,7], A8[8,11]
Huffman codes: {'a': '0', 'b': '111', 'c': '101', 'd': '100', 'r': '110'}
Encoded bits: 01111100101010001111100
Original size: 88 bits; encoded size: 23 bits
Encoded/original ratio: 26.14%
Charts saved: knapsack_allocation.png, activity_timeline.png, huffman_tree.png
```